SDS8102: Advanced Mathematical Statistics

Week 12 – Lecture 2
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4 Statistical properties of kernel estimators (contd.)

We finished the previous lecture by proving the following theorem.

Theorem 1. *Assume that the kernel K satisfies the assumptions of Propositions 1 and 2, also assume that $h = \alpha n^{-\frac{1}{2\beta+1}}$ for some positive α. Then the kernel density estimator $\hat{p}_n$ satisfies*

$$\sup_{x\in\mathbb{R}} \sup_{p\in\mathcal{P}(\beta,L)} \mathbb{E}((\hat{p}_n(x) - p(x))^2) \le C_* n^{-\frac{2\beta}{2\beta+1}}$$

where C_* is a constant that only depends on β, L, α and the kernel $K(\cdot)$.

Remarks:

- The pointwise bound in the theorem can be turned into a bound on the integral over a bounded interval $[a, b]$ with the same rate of convergence:

$$\sup_{p\in\mathcal{P}(\beta,L)} \mathbb{E} \int_a^b |\hat{p}_n(x) - p(x)|^2 dx \le \bar{C}_* n^{-\frac{2\beta}{2\beta+1}}$$

- The result of the theorem is only an upper bound. Is it optimal? The answer is yes. In fact, ws will show later that, for any density estimator T_n, we have

$$\sup_{p\in\mathcal{P}(\beta,L)} \mathbb{E}((T_n(x) - p(x))^2) \ge C_* n^{-\frac{2\beta}{2\beta+1}}$$

  for any fixed $x \in \mathbb{R}$, which means the rate we get from the theorem cannot be improved.

- How to choose h in practice? In the setup of the theorem, the value of optimal h depends on β and L. Is it possible to construct an estimator that has the same bound on the risk but is independent of β and L? Such an estimator would be called adaptive.

  The answer is as follows: if we are estimating under the risk MSE (at a fixed point), then it is only possible to achieve the rate $(\frac{\log n}{h})^{\frac{2\beta}{2\beta+1}}$ in an adaptive way. The $\log n$ term is the price to pay for adaptation, and one can show that it is unavoidable.

  However, if we consider the L_2-risk, then the rate $n^{-\frac{2\beta}{2\beta+1}}$ can be achieved adaptively. We will see later that this can be done by the cross-validation procedure.

- A kernel of order l can be constructed in multiple ways. For example, we can construct such a kernel based on orthonormal polynomials.

  Indeed, let $\{\phi_k\}_{k=0}^{\infty}$ be the orthonormal basis of Legendre polynomials on $[-1,1]$. Define kernel K to be

  $$K(u) = \sum_{m=0}^{l} \phi_m(0)\phi_m(u)I(|u| \le 1).$$

  Let us show that K is an order l kernel.

  Proof: Since the ϕ_k form an orthonormal basis of $L_2([-1,1])$, and ϕ_k is a polynomial with degree k, the function u^j could be represented, for any $j \le l$, by a linear combination of the form

  $$u^j = \sum_{s=0}^{j} b_s \phi_s(u),$$

  where b_s are some coefficients. This and the definition of K yield

  $$\int u^j K(u)du = \sum_{s=0}^{j}\sum_{m=0}^{l} b_s\phi_m(0) \int_{-1}^{1} \phi_m(u)\phi_s(u)du.$$

  The integral on the right hand side is just $\delta_{ms} = I(m = s)$, so we can get rid of the sum over m, and write

  $$\int u^j K(u)du = \sum_{s=0}^{j} b_s\phi_s(0) = 0^j = 1 \cdot I(j = 0) + 0 \cdot I(j \ne 0),$$

  which implies that K is an order l kernel. ■

  We may note that this kernel, in general, takes on negative values at some intervals.

- Notice that the rate of convergence $n^{-\frac{2\beta}{2\beta+1}}$ is much larger than n^{-1}. But as β goes to infinity(the function gets smoother and the class gets smaller), the rate approaches n^{-1}, which corresponds to the rate of convergence of the squared error in parametric estimation settings.

5 Fourier analysis of kernel density estimators

First introduce a measure of the risk $MISE$(mean integrated squared error)

$$MISE = \mathbb{E}\int (\hat{p}_n(x) - p(x))^2 dx \triangleq \mathbb{E}[\|\hat{p}_n - p\|_2^2].$$

From now on, assume that the density p is in $L_2(\mathbb{R})$ (then automatically $p \in L_2(\mathbb{R}) \cap L_1(\mathbb{R})$ since all densities are integrable).

Let us breifly review some facts from Fourier analysis.

Assume that $f \in L_1(\mathbb{R})$. Then the Fourier transform of f is

$$\mathcal{F}[f](w) = \int e^{iwx} f(x)dx.$$

Sometimes we will also use a shorter notation: $\mathcal{F}[f](w) = \hat{f}(w)$. For the Fourier transform of density p, we will use a special notation

$$\phi(w) = \int e^{iwx} p(x)dx$$

(this is the *characteristic function* associated to p).

If $f, \hat{f} \in L_2(\mathbb{R}) \cap L_1(\mathbb{R})$, then we have

- $\frac{1}{2\pi}\|\hat{f}\|_2^2 = \|f\|_2^2$ (Plancherel's equality).
- $f(x) = \frac{1}{2\pi}\int e^{-iwx}\hat{f}(w)dw$ (inversion of the Fourier transform).

The next question we are going to focus on is: How to define the Fourier transform for $f \in L_2(\mathbb{R})$? For example, the sinc kernel

$$K(u) = \frac{\sin u}{\pi u}$$

(with the value at 0 completed by continuity argument: $K(0) = \frac{1}{\pi}$) is in $L_2(\mathbb{R})$ but not in $L_1(\mathbb{R})$. With the definition we are going to give, its Fourier transform is

$$\hat{K}(w) = I(|w| \le 1).$$

The basic idea of the definition is to use the fact that $L_1(\mathbb{R}) \cap L_2(\mathbb{R})$ is dense in $L_2(\mathbb{R})$. Indeed, for $f \in L_2(\mathbb{R})$, we can construct the sequence of truncations of f:

$$f_m(x) = f(x)I(|x| \le m), m = 1, 2, ...$$

Then $f_m \in L_1(\mathbb{R}) \cap L_2(\mathbb{R})$, so we can define the Fourier transform of f_m in classical sense and denote it by $\hat{f}_m$. Moreover, f_m converges to f in the L_2 sense:

$$\|f_m - f\|_2^2 \to 0, n \to \infty.$$

Thus the f_m form a Cauchy sequence, i.e.,

$$\|f_m - f_{m'}\|_2^2 \to 0, m, m' \to \infty.$$

By Plancherel's equality,

$$\frac{1}{2\pi}\|\hat{f}_m - \hat{f}_{m'}\|_2^2 = \|f_m - f_{m'}\|_2^2 \to 0$$

Thus the $\hat{f}_m$ form a Cauchy sequence in $L_2(\mathbb{R})$. Since $L_2(\mathbb{R})$ is complete, the limit of $\hat{f}_m$ exists. We denote this limit by $\hat{f}$, and we call it the Fourier transform of f[1]. The properties of classical Fourier transform are preserved under this definition. In particular, since

$$\|\hat{f}_m\|_2 \to \|\hat{f}\|_2 \text{ and } \|f_m\|_2 \to \|f\|_2,$$

by taking limits of both sides of Plancherel's equality in classical sense for f_m and $\hat{f}_m$, we obtain:

$$\frac{1}{2\pi}\|\hat{f}\|_2^2 = \|f\|_2^2.$$

[1]Notice that $\hat{f}$ is defined up to an arbitrary modification of its values on a set of Lebesugue measure 0; strictly speaking, it is an equivalence class of functions.

Thus, the Fourier transform defined as above is an isometric isomorphism of L_2 spaces. One can show that if, in addition, $\hat{f} \in L_1(\mathbb{R})$, then

$$f(x) = \frac{1}{2\pi} \int e^{-iwx} \hat{f}(w) dw.$$

In particular, this relation is easily checked for the sinc kernel and its Fourier transform $I(|w| \leq 1)$.

We will need the following properties of the Fourier transform that are valid for both classical and L_2 definitions:

1. $\mathcal{F}[af(\cdot)](w) = a\mathcal{F}[f(\cdot)](w)$.
2. $\mathcal{F}[\frac{1}{h} f(\frac{\cdot}{h})](w) = \mathcal{F}[f(\cdot)](hw)$.
3. $\mathcal{F}[f(t-\cdot)](w) = e^{iwt}\mathcal{F}[f(\cdot)](-w)$.

Now let us find the Fourier transform of the kernel density estimator $\hat{p}_n$. By the definition of kernel density estimator and the linearity of the Fourier transform,

$$\mathcal{F}[\hat{p}_n(\cdot)](w) = \frac{1}{n} \sum_{j=1}^{n} \mathcal{F}\left[\frac{1}{h} K\left(\frac{X_j - \cdot}{h}\right)\right](w).$$

Using the properties 2 and 3 above, the right hand side is equal to

$$\frac{1}{n} \sum_{j=1}^{n} \mathcal{F}\left[\frac{1}{h} K\left(\frac{\cdot}{h}\right)\right](-w) e^{iwX_j} = \frac{1}{n} \sum_{j=1}^{n} e^{iwX_j} \hat{K}(-hw)$$

Consider

$$\phi_n(w) = \frac{1}{n} \sum_{j=1}^{n} e^{iwX_j},$$

that we will call the *empirical characteristic function*. Clearly, $\phi_n(w)$ is an unbiased estimator of the characteristic function $\phi(w)$.

From now on, we will always consider kernels that are symmetric:

$$K(u) = K(-u), \forall u \in \mathbb{R}.$$

Note that K is symmetric if and only if $\hat{K}$ is symmetric and real valued. To conclude, our final result is as follows: If K is symmetric, the Fourier transform of $\hat{p}_n$ is given by

$$\boxed{\mathcal{F}[\hat{p}_n(\cdot)](w) = \phi_n(w)\hat{K}(hw),}$$

where $\phi_n(w) = \frac{1}{n} \sum_{j=1}^{n} e^{iwX_j}$ is the empirical characteristic function.

More generally, for a symmetric function Λ in $L_2(\mathbb{R})$, we may define the following estimator

$$\hat{p}_{n,\lambda}(x) = \frac{1}{n} \sum_{j=1}^{n} \Lambda(X_j - x).$$

This is more general than the kernel estimator because if we take $\Lambda(x) = \frac{1}{h}K(\frac{x}{h})$, then $\hat{p}_{n,\lambda}$ becomes the kernel estimator. If we write $\mathcal{F}[\Lambda] = \lambda$, then the Fourier transform of $\hat{p}_{n,\lambda}$ is

$$\mathcal{F}[\hat{p}_{n,\lambda}](w) = \phi_n(w)\lambda(w).$$

Example: Consider the function

$$\Lambda(x) = C\frac{\sin((m+\frac{1}{2})x)}{\sin(x/2)}I(|x| \le \pi)$$

called the Dirichlet kernel. Here $C > 0$ is a normalizing factor, and m is an interger that plays the same role as the smoothing parameter h for kernel estimators.

Lemma 1. *The empirical characteristic function $\phi_n(w)$ satisfies:*

(i) $\mathbb{E}(\phi_n(w)) = \phi(w)$,

(ii) $\mathbb{E}(|\phi_n(w)|^2) = (1-\frac{1}{n})|\phi(w)|^2 + \frac{1}{n}$,

(iii) $\mathbb{E}\big(|\phi_n(w) - \phi(w)|^2\big) = \frac{1}{n}(1 - |\phi(w)|^2)$.

Proof: The first statement is obvious. To show (ii), write the left hand side as

$$\mathbb{E}(|\phi_n(w)|^2) = \mathbb{E}(\phi_n(w)\overline{\phi_n(w)}).$$

Recalling the definition of $\phi_n(w)$ we have $\overline{\phi_n(w)} = \phi_n(-w)$. So, the right hand side can be written as

$$\mathbb{E}\left(\frac{1}{n^2}\sum_{j,j'=1}^{n} e^{iw(X_j - X_{j'})}\right).$$

The contribution from the diagonal terms ($j = j'$) adds up to 1, and the off-diagonal terms can be factorized using independence of X_j's. Thus the expression above is equal to

$$\frac{1}{n} + \frac{1}{n^2}\sum_{j \neq j'} \mathbb{E}(e^{iwX_j})\mathbb{E}(e^{-iwX_{j'}}) = \frac{1}{n} + \frac{1}{n^2}(n(n-1))\phi(w)\phi(-w).$$

Notice that $\phi(w)\phi(-w) = |\phi(w)|^2$, and we have shown the desired equality. Finally, (iii) is a consequence of (ii). We expand the left hand side of (iii):

$$\mathbb{E}\big(|\phi_n(w) - \phi(w)|^2\big) = \mathbb{E}(|\phi_n(w)|^2) - 2\mathbb{E}(\phi_n(w)\overline{\phi(w)}) + |\phi(w)|^2$$

and we plug in result of (ii) to deduce (iii). ∎

Note that (iii) implies that $\phi_n(w)$ has a MSE with rate $1/n$.

Our aim now is to analyze the $MISE$ of generalized estimator $\hat{p}_{n,\lambda}$:

$$MISE = \mathbb{E}\int (\hat{p}_{n,\lambda}(x) - p(x))^2 dx = \mathbb{E}[||\hat{p}_{n,\lambda} - p||_2^2].$$

By Plancherel's equality,

$$\begin{aligned}\mathbb{E}[||\hat{p}_{n,\lambda} - p||_2^2] &= \mathbb{E}\Big[\frac{1}{2\pi}\int |\phi_n(w)\lambda(w) - \phi(w)|^2 dw\Big] \\ &= \mathbb{E}\Big[\frac{1}{2\pi}\int |(\phi_n(w) - \phi(w))\lambda(w) - \phi(w)(1-\lambda(w))|^2 dw\Big].\end{aligned}$$

Notice that the term involving $\phi_n(w) - \phi(w)$ plays the role of random deviation, and the other contributions are not random. Fubini's Theorem allows us to exchange the expected value and the integral. This gives

$$\begin{aligned}\mathbb{E}[||\hat{p}_{n,\lambda} - p||_2^2] = \frac{1}{2\pi}\Big[\int \{&\mathbb{E}|\phi_n(w) - \phi(w)|^2\lambda^2(w) + \\ &+ 2\mathbb{E}(\phi_n(w) - \phi(w))\lambda(w)(-\phi(w))(1-\lambda(w)) + |\phi(w)|^2(1-\lambda(w))^2\}dw\Big].\end{aligned}$$

Notice that the cross product term is the expected value of $\phi_n(w) - \phi(w)$ times a deterministic value. Since $\phi_n(w)$ is an unbiased estimator of $\phi(w)$, this term vanishes. Now apply part (iii) of Lemma 1 to obtain

$$MISE = \frac{1}{2\pi}\int \Big[\frac{1}{n}(1-|\phi(w)|^2)\lambda^2(w) + |\phi(w)|^2(1-\lambda(w))^2\Big]dw.$$

Thus, we have proved the following result.

Theorem 2. *Assume that $p \in L_2(\mathbb{R})$, and λ is a symmetric real-valued function in $L_2(\mathbb{R})$, then the $MISE$ of $\hat{p}_{n,\lambda}$ is*

$$\mathbb{E}[||\hat{p}_{n,\lambda} - p||_2^2] = \Big\{\frac{1}{2\pi}\int |\phi(w)|^2(1-\lambda(w))^2 dw + \frac{1}{2\pi n}\int \lambda^2(w)dw\Big\} - \frac{1}{2\pi n}\int |\phi(w)|^2\lambda^2(w)dw.$$

We will refer to the first and the second terms in the curly brackets as the bias squared term and the variance term respectively. The last term will be call the "bonus" term because it is negative and thus makes the error smaller. However, it is usually negligible as compared to the bias and variance terms.